Recent Magneto Inertial Fusion Experiments on the Field Reversed Configuration Heating Experiment

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Abstract. Magneto-Inertial Fusion (MIF) approaches take advantage of embedded magnetic field to improve plasma energy confinement by reducing thermal conduction relative to conventional inertial confinement fusion (ICF). MIF reduces required precision in the implosion and the convergence ratio. Since 2008 [1] and since our prior refereed publication on this topic [2], AFRL and LANL have developed further one version of MIF. We have (1) reliably formed, translated, and captured Field Reversed Configurations (FRCs) in magnetic mirrors inside metal shells or liners in preparation for subsequent compression by liner implosion; (2) imploded a liner with interior magnetic mirror field, obtaining evidence for compression of a 1.36 T field to 540 T; (3) performed a full system experiment of FRC formation, translation, capture, and imploding liner compression operation; (4) identified by comparison of 2D-MHD simulation and experiments factors limiting the closed- field lifetime of FRCs to about half that required for good liner compression of FRCs to multi-keV, 10^{19} ion/cm^3, high energy density plasma (HEDP) conditions; and (5) designed and prepared hardware to increase that closed field FRC lifetime to the required amount. Those lifetime experiments are now underway, with the goal of at least doubling closed-field FRC lifetimes and performing FRC implosions to HEDP conditions this year. These experiments have obtained imaging evidence of FRC rotation, and of initial rotation control measures slowing and stopping such rotation. Important improvements in fidelity of simulation to experiment have been achieved, enabling improved guidance and understanding of experiment design and performance.

1. Introduction

In Magneto-Inertial Fusion (MIF), one uses magnetic inhibition of electron thermal conduction to greatly reduce the required implosion velocity and density – radius product relative to un-magnetized inertial fusion concepts [3]. We report here on progress in developing the version of MIF that uses a Field Reversed Configuration (FRC), formed by a reversed field theta discharge, and its subsequent compression by an imploding metal shell or liner, which is driven by a Z-pinch geometry discharge [3,4]. We previously reported on the successful demonstrations of imploding aluminum liners with size, radial convergence, uniformity, and implosion velocity suitable for compressing an FRC [5], and on the use of deformable liner – electrode contacts, which enable the use of large electrode apertures, suitable for FRC injection [2,6]. Our

colleagues from Los Alamos (co-authors in this paper and authors and co-authors in [1-8]) and we have reported on formation of FRC's at the Los Alamos FRX-L facility, which have density ~ 5 x10^{16} cm^{-3}, ion + electron temperature ~ 400 eV, flux exclusion radius ~ 2 cm, and lifetimes ~ 10 μs [3,7]. We have reported on the essential reproduction of such a facility and its experimental results adjacent to the AFRL Shiva Star capacitor bank, which is used to drive liner implosions [8]. We have also previously reported two dimensional magneto-hydrodynamic (2D-MHD) simulations of FRC formation, translation to liner interior, and FRC compression to multi-KeV temperatures [5, 6]. More recent progress includes performing liner implosions with magnetic mirror fields interior to the liner; an engineering test of the full system including mirror fields, FRC formation translation, and capture, with liner implosion operation; an experimental series to improve captured FRC lifetime by improving flux capture using RF assisted pre-ionization, optimizing the relative timing of the several capacitor discharges to form the FRCs, and by introducing rotation control using biased rings and gas puffing to enable the biased ring operation. Recent progress also includes improved 2D-MHD simulation fidelity to magnetic probe and interferometry signals in experiments on FRC formation and capture. Prior reports by a number of researchers suggest the general concept of using liners to compress plasmas and research on shorter or lower velocity liners [9-22], and implosion of a Cu-W liner with explosives to compress flux to 200 T [23]. Present related work includes the Sandia MagLIF (Magnetized Liner Implosion Fusion) effort [24]. This MagLIF effort uses open field lines, and solid density liners which vaporize early in the implosion, with 100 to 200 nanosecond implosion time. In our FRCHX effort, we use closed poloidal field lines, with much larger solid liners that remain solid and retain material strength through most of the implosion, which lasts about 20 microseconds. Present related work on lower density, larger sized FRCs is being done by Tri Alpha Energy, which is using plasma guns as an alternative means of ionization and end shorting control, and which uses two opposed reversed field theta discharges to merge two FRCs [25].

2. Experiment Description

The experiment hardware is shown in Fig. 1. The FRC is formed using a reversed field theta discharge, employing 6 capacitor bank discharges to form, guide, and trap the FRC. The FRC formation begins with a 130 μs rise discharge establishing a ~ 0.4 T axial bias field. Cusp fields with 40 μs rise enhance magnetic field reconnection.

The FRC is formed in a 50 to 100 milliTorr deuterium gas prefill or gas puff inside a 10 cm diameter quartz tube inside the segmented theta coil. The coil is segmented for diagnostic access. The gas is pre-ionized using a 1 μs rise time ringing theta pinch geometry discharge. Ideally, the plasma formed at this point would be fully diffused in the axial bias field. Then a 3.2 to 3.3 μs rise, ~ 1.5 to 2 T reverse bias axial field is applied to complete FRC formation. The theta coils feature a conical bore to propel the FRC into the guide and mirror regions of the apparatus. The guide and mirror coils are energized by a relatively slow 4.7 ms rise capacitor bank. The guide and mirror field inside the Al liner is formed by an array of coils downstream of (above) the formation region, with the mirror field inside the Al liner. This field is diffused through the liner

implosion hardware and related structure. An example of the axial profile of this predominantly axial guide and mirror field is shown in Fig. 2, with bias and cusp field contributions also shown. Large area inductive magnetic probe measurements and 3D magnetic diffusion calculations using COMSOL [26] agree substantially on this field profile. This field is diffused through the liner implosion hardware and related structure. The 30 cm long x 10 cm diameter x 0.11 cm thick liner implosion is driven by a 4.5 MJ, 12 MA, 85 kV discharge produced by the AFRL Shiva Star capacitor bank.

Diagnostics include pulsed power current and voltage, magnetic probes, flux loops, fast multi-frame imaging, flash radiography, optical photodiodes, Vacuum Ultra-Violet and X-ray diodes, optical spectroscopy, neutron activation, BTI bubble, and scintillator photomultiplier tube detectors.

3. Rotation Control Theory

Prior to discussing results, it is useful to discuss some rotation control theory to explain our motivation for using biased rings, which are discussed in the next section.

Assuming high electrical conductivity everywhere, the generalized Ohm's law's ∇p_e contribution to the **E** field implies that an FRC exterior that is not rotating at the midplane (azimuthal ion diamagnetic flow and **E** × **B** drift cancel each other out there) will have an **E** field crossing the **B** field at the wall wherever the external open **B** field lines exits the chamber [27]. Whether the wall is dielectric or (especially) metal, the conductivity of the wall or plasma layer there will carry current and eventually short out this **E** field. This results in a lowering of the radial **E** field at midplane too, so the FRC exterior starts to rotate in the ion diamagnetic flow direction. The FRC interior will then rotate too due to viscous coupling. This leads to centripetal acceleration-driven modes, for which $n = 2$ is the first that the gryoviscous stress tensor can no longer stabilize. This occurs once the rotational speed reaches the order of the ion diamagnetic flow speed [28]. In a brief final comment in his paper, Aso [29] suggests supporting the **E** field to prevent rotation. The principle of operation of the biased rings in our experiment is to do so by passing open magnetic flux as it exits the chamber (radially, above the highest **B** field coil, in our system) between conducting rings connected to an external capacitor bank or banks charged so as to maintain the midplane radial **E** field at a level sufficient to prevent unstable rotation rates [30]. Steinhaur [27] provides the theoretical basis from which we estimate the voltage requirements for doing so. Substituting his Eq. 7 (electric potential at a given radius in the midplane that results from an assumed isothermal integration of $E_\parallel = \nabla_\parallel p_e / e n_e$ along the open **B** field lines from there to the wall) into his Eq. 2 (midplane ion diamagnetic flow plus **E** × **B** drift velocities) solved for zero (no rotation), one may solve for the required wall potential by integration,

$$\phi_0(\psi) = \frac{kT_e(r_s)}{e}\ln\frac{n_e(r_s)}{n_{e0}(0)} - \frac{kT_e(r)}{e}\ln\frac{n_e(r)}{n_{e0}(\psi)} - \int_{r_s}^{r}\frac{1}{q_i n_i(r')}\frac{dp_i(r')}{dr'}dr' + \phi_0(0) \quad (1)$$

Here, r is the radius specifically at the midplane, ψ is the axial magnetic flux interior to r at the midplane, r_s is the midplane seperatrix radius (where $\psi = 0$), and zero subscripts refer to properties at the wall connected by a **B** field line passing through radius r at the midplane (enclosing and, therefore, parameterized by flux ψ). If we assume, for the purposes of rough estimation of ring voltage requirements, that $T = T_e = T_i$ are uniform, the plasma is singly ionized, and that $n = n_e = n_i$, this expression simplifies to,

$$\phi_0(\psi) = \frac{kT}{e}\ln\frac{n_0(\psi)n^2(r_s)}{n_0(0)n^2(r)} + \phi_0(0) \quad (2)$$

This indicates that, in volts, only a few times the plasma temperature in electron volts is required.

4. Experimental results on liner implosions and FRC capture

Liner implosion experiments with an operationally relevant interior magnetic mirror field obtained evidence for compression 1.36 T field to 540 T, by comparing liner inner and outer radius timing from flash radiography with MHD simulation, as illustrated in Fig.3.

Radiographs for this compression experiment are shown in Fig. 4. We used two 300 kV, 5 KA, 30 nanosecond radiography pulsers, operated at 250 kV, with demountable, in-house built radiography tubes. The bremsstrahlung spot size is small, about 0.1 cm, based on empirical resolution. We used digital film, which has several advantages, including large dynamic range, the acquisition of the radiographs in already digital form, and the ability to retrieve data from film even if substantially damaged, as was the case for the earlier radiograph, despite the need to image through 6 cm of polyethylene for both fragment shielding and structural mounting of magnetic mirror field coils. The field of view was limited by the magnetic mirror field coils to partial axial coverage. The radiography exposure was also modulated by the use of a slotted 0.6 cm thick Al outer conductor for the liner implosion discharge. This slotted feature is necessary to insert the mirror field into the interior when using a 4.7 ms rise time discharge to energize the mirror coils.

A full engineering systems test which exercised the FRC formation discharges and liner implosion was successful in the pulsed power operations sense, but the signatures of plasma compression were weak, which was expected because captured closed magnetic field FRC lifetimes were not yet what we believe are needed (about 20 μs, depending on relative timing of liner implosion discharge and FRC reversed field formation discharge). A subsequent experimental campaign, using internal magnetic probes in the liner-mirror region, has confirmed FRC capture, but with insufficient closed field lifetime to support compression. We are now implementing measures to increase that lifetime. One approach uses a two-stage pre-ionization to enhance trapped bias flux. The first stage uses RF to enable better flexibility in the timing of the

second stage pre-ionization and the reverse field discharge. This two-stage pre-ionization in which the RF electric fields are axial, achieved using a pair of ring electrodes outside the quartz tube above and below the formation region, appears to result better flux capture and larger FRC exclusion radius. These measures also include using charged rings above the liner/mirror region to control end shorting as a means to reduce or stop life-limiting FRC rotation. Use of the rings, requires replacing the approximately 50 milli-Torr deuterium gas prefill with a gas puff, if one needs to use ring voltages above 200 volts.

The RF first stage ionization uses ring electrodes, shown in Fig. 5. Two Cu foil ring electrodes are mounted in grooves in the inner surface of nylon rings which surround the 11 cm outer diameter quartz tube of the FRC formation hardware, in the lower and upper parts of the FRC formation region. The lower ring is visible in Fig. 5. The center conductor of a co-axial cable is connected to the outer surface of the lower RF ring electrode and the ground of the same co-axial cable is connected to the upper RF ring electrode. Low-amplitude, long-duration (250 ~ 500 V, ~ 10 ms, 46.79 MHz, KW range) or large-amplitude, short-duration (~30 kV, 62 ~ 125 ns, 32 MHz, MW range) axial electric fields (parallel to the bias magnetic field) are applied between the two rings. (The frequencies are due mainly to available equipment.) Either version of RF first stage ionization, slow or fast, appears to enable ionization of the prefilled (or puffed) gas prior to complete nulling of the bias field by the fast theta pre-ionization discharge, which should improve flux capture.

An FRC, once formed, starts to rotate, as discussed in the previous section. Therefore, we use biased rings to slow or prevent rotation. Fig. 6 illustrates the biased rings used to attempt control of such rotation. These rings are inserted just above (downstream of) the top of the solid liner/upper implosion electrode/top of magnetic trap region. Figs, 7, 8, and 9 illustrate the gas puff hardware, puffed gas density vs time and position, and the fact that using gas puffing rather than gas prefill results in similar flux exclusion radius vs time in the FRC trapping region. The FWHM of this exclusion radius vs time, if large enough, is a measure of the closed field lifetime.

Some examples of magnetic field signals and exclusion radii in the capture region, in the magnetic mirror well inside the liner, obtained using biased rings (but no RF first stage ionization), are shown in Fig. 10. The excluded radius is obtained from the magnetic probe data using, for example, the Tuszewski method [31]. The maximum exclusion radius FWHM, a measure of lifetime in the trapped region, is in these examples 11.5 μs for 800 volt ring bias. When no biased rings are used, and either no RF first stage ionization or long duration low power (kilowatt range) RF first stage ionization is used, the exclusion radius FWHM is typically 8 to 9 μs. In practice, it is found that the high electrical conductivity conditions assumed by the analysis in the previous section on rotation control theory occur too late at the wall if one depends exclusively on plasma leakage from the FRC to close the bias control circuit. Using a small gas prefill or adjusting the gas puff distribution enables earlier closing of the bias control circuit, but we believe that this will also require lower inductance bias ring connections to achieve rotation control prior to FRC decay. Fig. 11 shows 100 ns gated images, viewed axially from below. The earlier photos (4 frames on the left), for a case with higher flux capture and higher bias ring voltage, show intact, symmetric luminosity for the first 4 μs after probe luminosity indicates entrance into trapping region. By 7 μs after entrance, the plasma has moved toward one side – perhaps indicating wobbling. The later photos in Fig. 11 (on right, similar parameters except bias ring) provide an example of observed plasma rotation both with (left) and

without (right) a bias voltage applied to the rings. However, this is after the FRC has deteriorated or decayed. This indicates the need for earlier rotation control.

Axial view vacuum ultraviolet images using a soft x-ray pinhole snout coupled to a fluorescer and observed by the multi-frame Hadland camera, confirm the hollow profiles expected of an FRC [32]. With time resolved visible spectroscopy (390-510 nm range), we observe clean (Stark-broadened) deuterium Balmer series lines, with trace Oxygen impurity, in the theta coil formation region. By the time the translating FRC reaches the capture region, we have evidence of plasma wall or plasma – interior probe interactions, bringing in additional impurities. An example is shown in Fig. 12. This needs further study, including repeating without interior magnetic probes, with radial spatial resolution, and in VUV, to see if the central region remains low in impurity.

More recently, we have obtained improvements in this measure of trapped lifetime reaching these levels without the use of RF and bias rings, when increasing the firing delay between the fast theta discharge used for ionization and the main theta reversed field discharge. We are now combining this with RF first stage ionization and a faster response bias ring system.

5. Improved fidelity of simulations to experiment

Experiment design and interpretation of data have been strongly coupled to and guided by 2D and 3D-MHD and extended MHD simulations. The examples shown here are 2D simulations of experiments with both fixed (static, un-imploded) and imploding aluminum liners, performed using MACH2 [33]. These simulations include SESAME table equations of state and resistivities for materials, all three components of the magnetic fields, an energy-conserving model of the imploding deformable liner, and flux-based field modeling of the time-dependent poloidal fields. They use adaptive gridding for the FRC and a Lagrangian grid for the liner.

We have attempted to capture as much of the physical geometry as we can in our simulations. See Fig. 13. The improved diagnostics of our recent static liner experiments have provided a lot of data that can be used within the simulations and as a basis for validation of the simulation's representation of the experiment.

Prior 2D-MHD simulations of FRC formation, capture, and compression [2,8], using initially warm FRCs (0.5 eV temperature), predict optimistic FRC flux capture, and, assuming no rotation, compression of such FRCs to 8 KeV ion temperature and DD neutron yields in the 10^{12} to 10^{14} range. However, such simulations also predict capture region magnetic probe signals two or more times the amplitude and with wider pulse widths than those seen experimentally. Using the 'time of first light' in the experimental data, an indicator of when the ionization and flux-trapping occur, as the start time for such simulations produces integrated B-dot signatures in the capture region that are more similar to those measured in the corresponding experiments, as seen in Fig. 14.

The particular comparison shown in Fig.15 is based on an experiment performed in March of 2012, using low level RF-induced pre-pre-ionization, also known as RF first stage ionization. As seen in Fig.16, only part of the FRC was captured in the mirror fields in the static liner − some mass and flux escaped out the top (left hand side in picture) and the FRC split between part in the intended trapping region and part below that (right hand side in picture). This is also evident for the case shown in Fig. 17.

The imploding liner experiments have the added complication of timing the main theta bank of the FRC relative to the main bank of the liner implosion. In order to maximize the lifetime of the FRC within the imploding liner, it is useful to have the liner accelerated and imploding when the FRC arrives, this means the compression of the mirror fields at FRC insertion by the liner must be accounted for.

These simulations have shown that the amount of flux trapped during the pre-ionization and hence in the initial FRC affects the lifetime and the robustness of the FRC. We performed a similar compression simulation with data from another March 2012 experiment with higher RF and higher bias field at the time of first light. This FRC had 67% more flux after its formation than the FRC discussed above (1.257 milli-Weber (mW) versus 0.754 mW). The imploding liner captured more of this FRC (see Fig. 17) and the captured portion had more flux (0.415 mW vs 0.346 mW), was hotter (273 eV vs 209 eV), and was less dense (N_i = 2.6x10^{16} /cc vs 3.3x10^{16} /cc) than the low-RF FRC discussed previously. These features may improve its lifetime. Its yield upon compression to a 5 mm liner radius was increased by a factor of 25. This simulation calculated a yield ~2x10^{10} up to that point (with more likely as compression continues), with temperature 4 KeV, density ~10^{18} ions/cc, and length ~3 cm.

Analytic estimates of uncompressed, captured FRC DD neutron yields, based on low energy, low temperature cross sections and reactivities from [34], and assuming 3x10^{16} ions/cc density, 5 cm diameter, 15 cm length, and 10 μs duration, give results of ~ 10^3 for 0.2 KeV temperature, and 5x10^4 for 0.3 KeV temperature. This is below our measurement threshold, which is ~ 10^6 for FRC capture experiments, and a few x 10^7 for compression experiments (due to using greater detector distance and blast shielding for detector survivability). Such estimates for compressed FRCs range from ~10^9 to ~10^{14}, depending on assumed compressed density (10^{18} to few x 10^{19} ions/cc), temperature (1 to 8 KeV), and radial compression ranges from 10 to 20, with assumed peak compressed duration of 100 nanoseconds.

6. Future Work

An alternative to RF and bias rings which we are preparing to explore is to use a plasma gun for ionization and end shorting control. This has been employed, more than doubling the lifetime of lower energy density FRCs, as documented in the reports of Tri-Alpha Energy [25]. Future work will include FRC compression and study of FRC behavior and the liner inner surface during

compression, which is in a warm dense matter state. These effects include, e.g., Rayleigh Taylor effects and liner material - plasma mixing under HEDP conditions.

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8. References

[1] WURDEN, G.A, INTRATOR, T.P. et al, LA-UR-08-0796, "FRCHX Magnetized Target Fusion HEDLP Experiments **,"** IC/P4-13, IAEA 2008 Fusion Energy Conference, Geneva, Switzerland, Oct. 13-18, 2008.

[2] DEGNAN, J.H. et al, "Experimental and Computational Progress on Liner Implosions for Compression of FRCs," Transactions on Plasma Science 36, Part 1, p. 80, Feb 2008.

[3] SCHOENBERG, K.F., SIEMON, R.E. et al, "Magnetized Target Fusion, A Proof of Principle Research Proposal," LA-UR-98-2413, 1998

[4] TACETTI, J. M. Taccetti et al, Rev. Sci. Instr. 74, 4314 (2003).

[5] DEGNAN, J.H. et al, "Implosion of Solid Liner for Compression of Field Reversed Configuration," IEEE Transactions on Plasma Science 29, p.93-98 (2001).

[6] DEGNAN, J.H.et al, "Progress in Liner Implosions for Compression of FRC's," in Proceedings of the 200 6 International Conference on Megagauss Magnetic Fields and Related Topics, p. 93, Ed. G. F. Kiuttu, R. E. Reinovsky, and P. J. Turchi, IEEE Catalog Number CFP06MEG-PRT, 2007.

[7] ZHANG, S. Y. et al, "High density field reversed configuration plasma for magnetized target fusion," IEEE Transactions on Plasma Science 34, no. 2, p. 223, April 2006.

[8] DEGNAN, J.H.et al, "Research on Field Reversed Configuration (FRC) formation and their compression using liners," in Proceedings of Megagauss 12 Conference (2008)

[9] LINDEMUTH, I., KIRKPATRICK, R., "Parameter space for magnetized fuel targets in Inertial Confinement Fusion," Nucl. Fusion 23, 263 (1983)

[10] LINDEMUTH, I. et al., "Target plasma formation for Magnetic Compression/Magnetized Target Fusion," Phys.Rev.Lett. 75, 1953 (1995)

[11] CHERNYSHEV, V.K., MOKHOV, V. N., "On the progress in the creation of powerful explosive magnetic energy sources for thermonuclear target implosion," in Digest of Technical Papers: 8th IEEE International Pulsed Power Conference (IEEE, NY, NY, 1991), P.395-410

[12] FOWLER, C. M., “Intense Magnetic Fields” in Atomic Energy Research in the Life and Physical Sciences, 1960, (AEC Special Report, Jan. 1961), p.104.

[13] LINHART, J. G., KNOEPFEL, H., GOURLAN, C., “Amplification of magnetic fields and heating of plasma by a collapsing metallic shell,” Nucl.Fusion 1962 Supplement 2:733.

[14] WINSOR, N. K., BORIS, J. P., SHANY, R. A., “Plasma heating by flux compression,” Proceedings, Symposium on Plasma Heating and Injection, Varenna, Italy (Editorice Compositori, Bologna, 1973), p.227-231.

[15] VELIKHOV, E. P. et al, “Generation of megagauss magnetic fields using a liner compressed by high pressure gas,” Sov.Phys.-Tech.Phys.18, p.274 (1973).

[16] GOLOVIZNIN, V. et al, “Numerical simulation of dynamics of quasispherical metallic liner,” in “Megagauss Physics and Technology,” Second International Conference on Megagauss Magnetic Field Generation and related Topics, edited by P.J.Turchi (Plenum Press, New York, NY, 1980), p. 415

[17]TURCHI, P. J. et al, “Review of the NRL Liner Implosion Program,” *ibid*, p.395

[18] SHERWOOD, A. E. et al, “Results from the Los Alamos Fast Liner Experiment,” *ibid*, p. 391

[19] KAREEV, Ju. A. et al, “Liner thermonuclear system with superhigh magnetic field and $\beta>1$,” *ibid*, p. 399

[20] ANDRIANOV, A. M. et al, “Production of megagauss magnetic fields by compression of a magnetic field with a metallic liner,” *ibid*, p. 479

[21] MOKHOV, V. et al, “Possible solution of the controlled thermonuclear fusion problem based on magnetogasdynamic energy storage,” Sov.Phys. Dokl. 24, 557 (1979)

[22] CHERNYSHEV, V. K. et al, “Cylindrical liner: implosion dynamics under EMG magnetic pressure,” in Sixth International Conference on Megagauss Magnetic Field Generation and Related Topics, edited by M.Cowan and R.Spielman (Nova Science Publishers, Inc, Commack, NY, 1994), p.815

[23] ARMSTRONG, W. T., MORGAN, J. A., “Liner compression of magnetically confined FRC plasmas,” in "Megagauss Technology and Pulsed Power Applications," Fourth International Conference on Megagauss Magnetic Field Generation and related Topics, edited by C.M.Fowler, R.S.Caird, and D.J.Erickson (Plenum Press, New York, NY, 1987), p.683

[24] S. Slutz et al, “Pulsed-power driven cylindrical liner implosions of laser pre-heated fuel magnetized with an axial field,” Phys. Plasmas 17, 056303 (2010).

[25] GUO, H. Y.; BINDERBAUER, M. W. et al., "Formation of a long-lived hot field reversed configuration by dynamically merging two colliding high-β compact toroids," Physics of Plasmas, vol. 18, no.5, 56110 (2011).

[26] COMSOL Multiphysics R : www.**comsol**.com/papers/7149/download/Grunbaum.pdf

[27] STEINHAUER, L. C., "End-Shorting and electric field in edge plasmas with application to field-reversed configurations," *Phys. Plasmas* **9**, pp. 3851, 2002.

[28] RUDEN, E. L., "The FRC's n=2 rotational instability interpreted as the dominant Rayleigh-Taylor mode of a gyroviscous plasma with sheared toroidal flow," *US-Japan exchange 2004: New directions and physics for compact toroids*, Santa Fe, NM, Sept. 14-16, 2004.

[29] Y. Aso, W. Ch., H. S., and H. K., Nucl. Fusion 22, 843 (1982).

[30] E. L. Ruden and M. Frese, Bull. Amer. Phys. Soc. (Prog. Ann. Meet. DPP) 52, 1207 (2007).

[31] TUSZEWSKI, M., Phys. Fluids 24, p. 2126, 1981; TUSZEWSKI, M., ARMSTRONG, W. T., Rev. Sci. Instr. 54, p. 1611, 1983

[32] WURDEN, G. A. and COFFEY, S. K., "A multi-frame soft X-ray pinhole imaging diagnostic for single-shot applications," Rev. Sci. Instr. 83, 10E516 (2012)

[33] PETERKIN, Jr, R. E., FRESE, M. H., SOVINEC, C. R., "Transport of magnetic flux in an arbitrary co-ordinate ALE code," J.Comp.Phys. 140, 148 (1998)

[34] H. S. Bosch and G. M. Hale, "Improved Formulas for Fusion Cross-Sections and Thermal Reactivities," Nuclear Fusion, Vol. 32, No. 4, p.611-631 (1992).

FIGURE CAPTIONS

Fig. 1: Illustration and photo of FRC formation – capture - compression load under center of liner implosion Shiva Star capacitor bank. Left shows illustration of hardware design. Letters C through J indicate gaps in segmented single turn theta coil that allow diagnostic access for magnetic probes, flux loops, photodiodes, and interferometry lines of sight. Symbols T0 and T2 through T6 show axial locations of axial magnetic probes. Middle shows 2D-MHD generated field lines at a times, left to right, shortly after FRC formation, during FRC translation, and during compression by liner implosion after FRC capture inside the liner.

Fig.2 Axial guide and mirror field measured by large area probes at 6.1 ms after guide and mirror field discharge, with bias and cusp field contributions also shown.

Fig.3 MHD calculations and radiography data for a liner-on-field compression experiment.

Fig. 4 Radiographs from compression experiment. Top left is setup for early radiograph. Top right is dynamic early radiograph. Bottom left is setup for later radiograph. Bottom right is dynamic later radiograph. Radiograph times are 21.813 and 22.797 μs after start of liner implosion discharge. Liner outer diameter is 10 cm in setups. Dynamic radiograph liner outer diameters are 3.15 cm and 3.73 cm for early and late radiographs, respectively.

Fig. 5 Left photo shows lower of two RF ring electrodes (1.9 cm wide Cu foil on inner surface of nylon ring) surrounding 10 cm diameter quartz tube in FRC formation region. The two thinner nylon rings above are to mount flux loop diagnostic probes. Upper RF ring electrode is above

this field of view. Photo was taken during assembly, and shows half segments of theta discharge coil. Right photo shows 1.9 cm gap in Cu foil RF ring electrode. Kapton tape over one end of foil and under the other inhibits breakdown between the two ends. The center conductor of a co-axial cable is connected to the outer surface of the lower RF ring electrode (see red lead in lower right of left photo) and the ground of the same co-ax cable is connected to the upper RF ring electrode (not shown here).

Fig. 6 Illustration of bias rings inserted just above (downstream of) the top of the solid liner/upper implosion electrode/top of magnetic trap region. Open field lines exit through the inner conductor in this axial region. See section 3. Conducting leads to bias rings as well as co-axial leads to axial magnetic probes interior to the liner, with vacuum feed-throughs from the top are also shown. The interior probes are used in FRC capture experiments without subsequent liner implosions, but not in implosion - compression experiments.

Fig. 7 On left is shown partial illustration of FRC formation, translation, capture hardware (the vacuum boundary portion), next to illustration of gas puff test hardware, with 6AH6 probe locations indicated for 6 probes. On the right is shown a photo of the gas puff test hardware.

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Fig 10 Magnetic field signals (left) and exclusion radii (right) in the capture region, using biased rings for attempted rotation control, but no RF first stage ionization. Horizontal scales are time, 5 μs/div. Vertical scales are .05 Tesla/div and 0.5 cm/div.

Fig.11 Numbers adjacent to axial view images are times in μs after pre-ionization bank firing for all photos. The left set of 4 images, for case with more flux capture and higher bias ring voltage, show intact, symmetric luminosity until 16 μs, which is 4 μs after probe luminosity indicates entrance into trapping region. By the 4th frame (19 μs = 7 μs after entrance), the plasma has moved toward one side – perhaps indicating wobbling. The right set of images indicate rotation of a decayed FRC remnant with (left) and without (right) biased ring rotation control. Four bright spots are due to probes in capture region.

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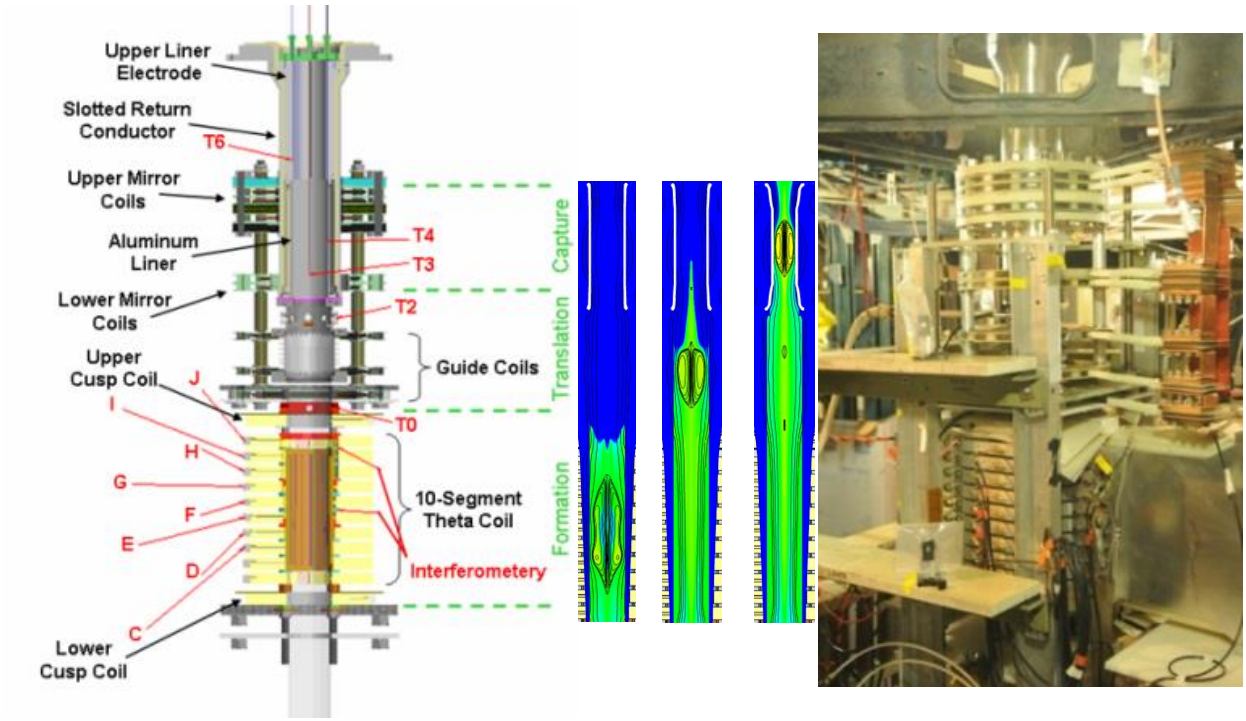


Fig. 1: Illustration and photo of FRC formation – capture - compression load under center of liner implosion Shiva Star capacitor bank. Left shows illustration of hardware design. Letters C through J indicate gaps in segmented single turn theta coil that allow diagnostic access for magnetic probes, flux loops, photodiodes, and interferometry lines of sight. Symbols T0 and T2 through T6 show axial locations of axial magnetic probes. Middle shows 2D-MHD generated field lines at a times, left to right, shortly after FRC formation, during FRC translation, and during compression by liner implosion after FRC capture inside the liner.

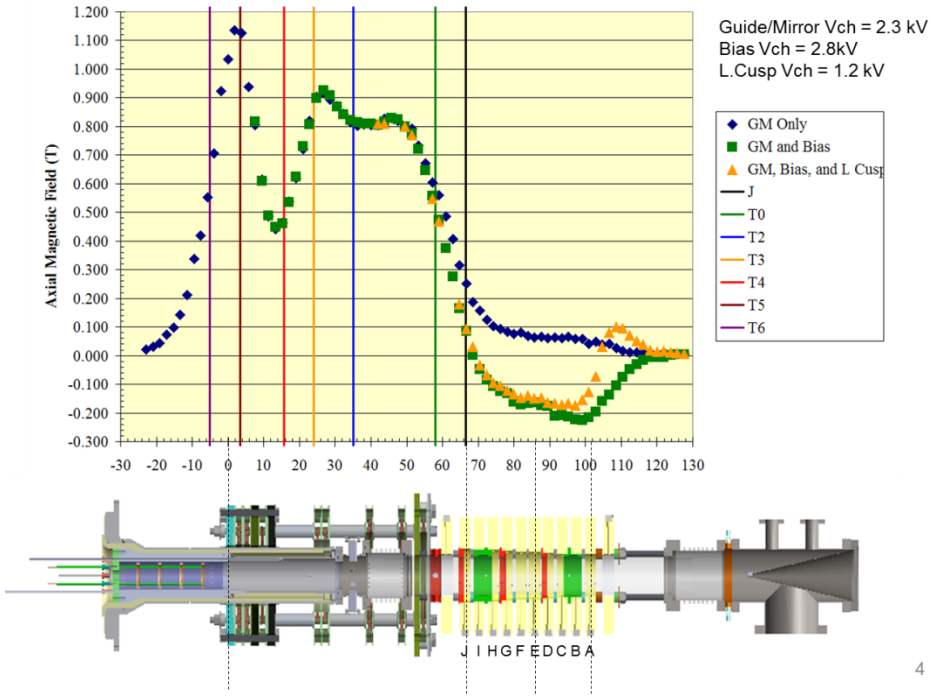


Fig.2 Axial guide and mirror field measured by large area probes at 6.1 ms after guide and mirror field discharge, with bias and cusp field contributions also shown.

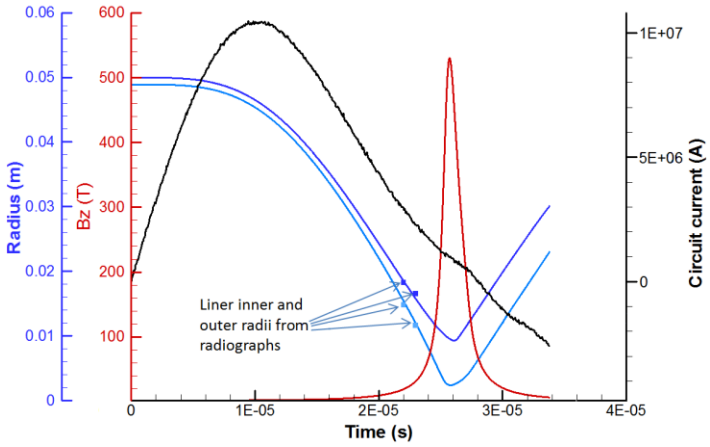


Fig.3 MHD calculations and radiography data for a liner-on-field compression experiment.

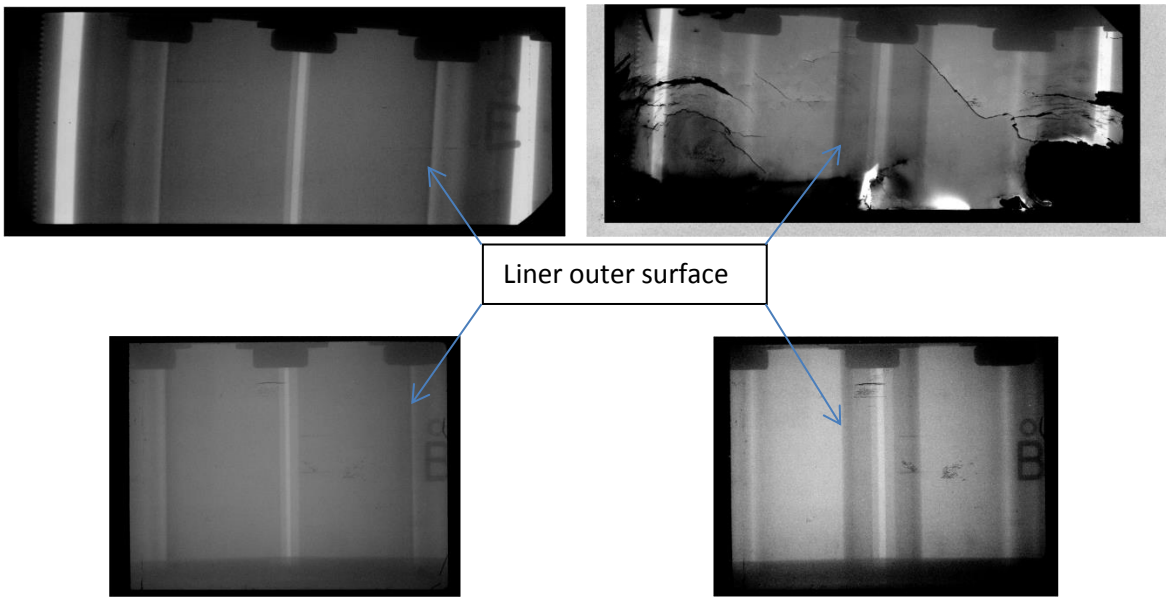


Fig. 4 Radiographs from compression experiment. Top left is setup for early radiograph. Top right is dynamic early radiograph. Bottom left is setup for later radiograph. Bottom right is dynamic later radiograph. Radiograph times are 21.813 and 22.797 μs after start of liner implosion discharge. Liner outer diameter is 10 cm in setups. Dynamic radiograph liner outer diameters are 3.15 cm and 3.73 cm for early and late radiographs, respectively.

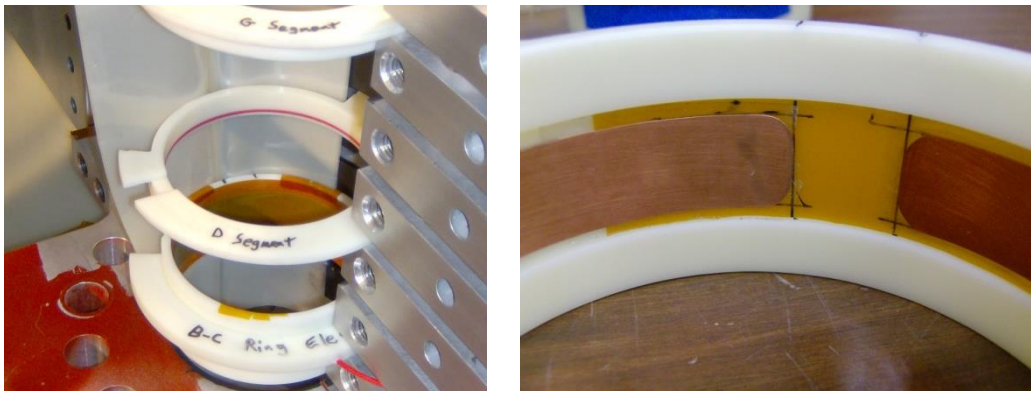


Fig. 5 Left photo shows lower of two RF ring electrodes (1.9 cm wide Cu foil on inner surface of nylon ring) surrounding 10 cm diameter quartz tube in FRC formation region. The two thinner nylon rings above are to mount flux loop diagnostic probes. Upper RF ring electrode is above this field of view. Photo was taken during assembly, and shows half segments of theta discharge coil. Right photo shows 1.9 cm gap in Cu foil RF ring electrode. Kapton tape over one end of foil and under the other inhibits breakdown between the two ends. The center conductor of a co-axial

cable is connected to the outer surface of the lower RF ring electrode (see red lead in lower right of left photo) and the ground of the same co-ax cable is connected to the upper RF ring electrode (not shown here).



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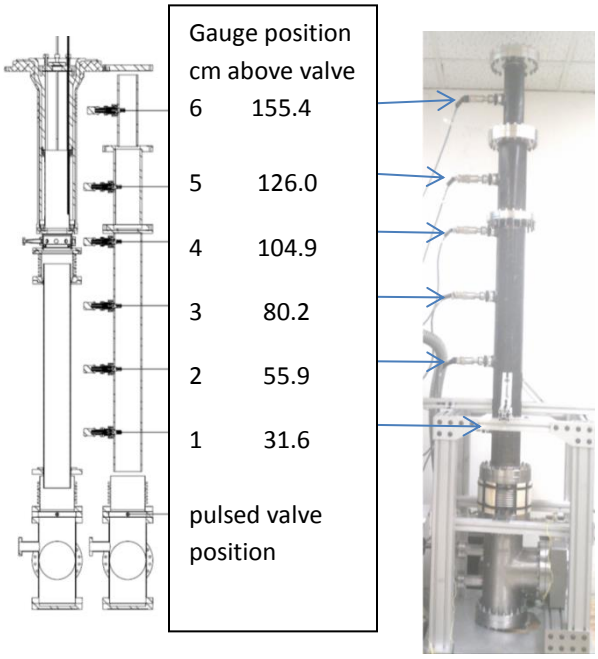


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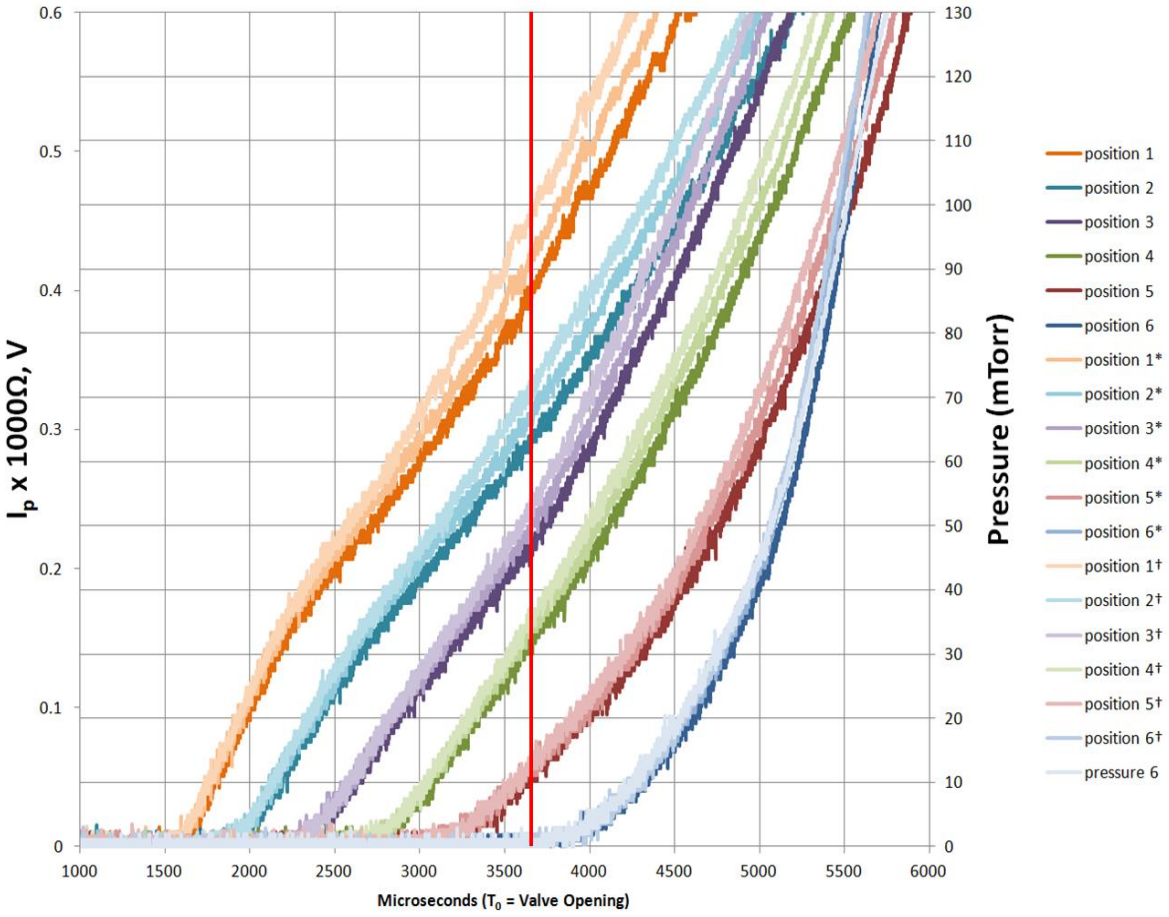


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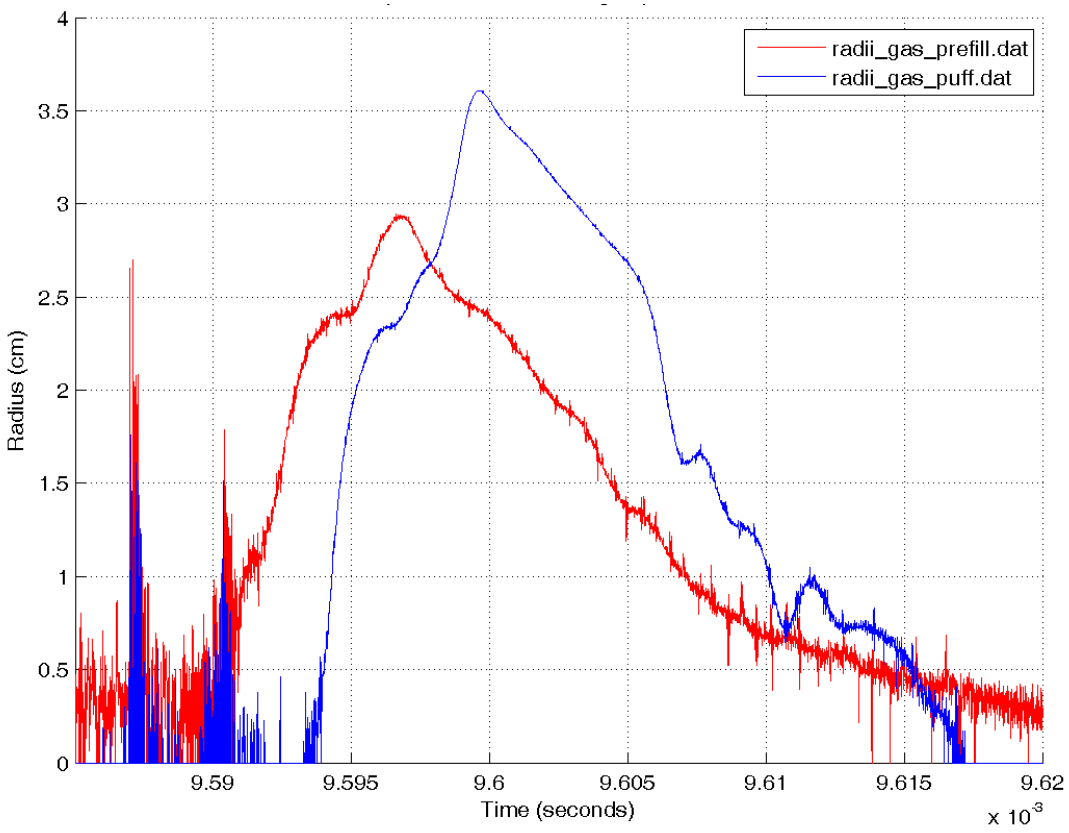


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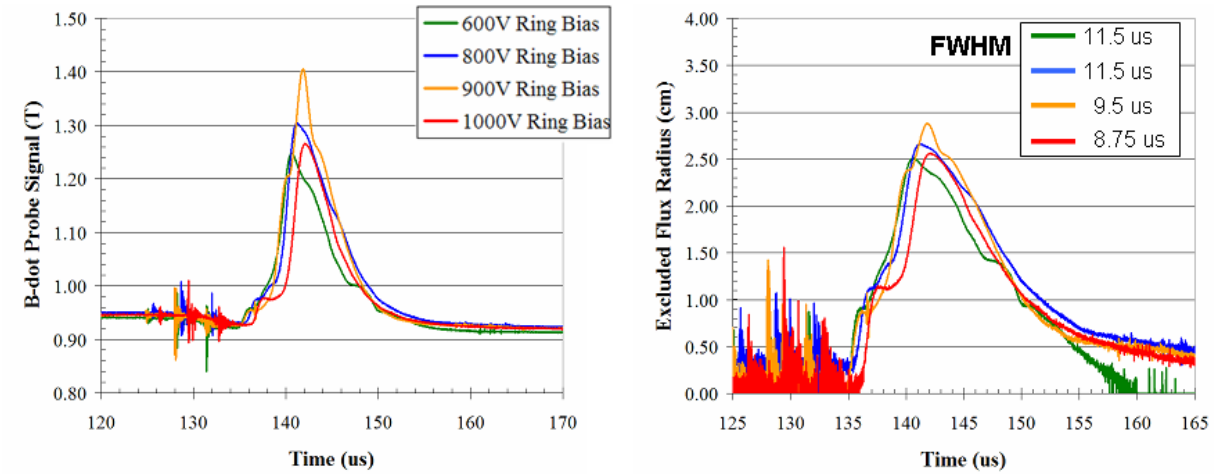


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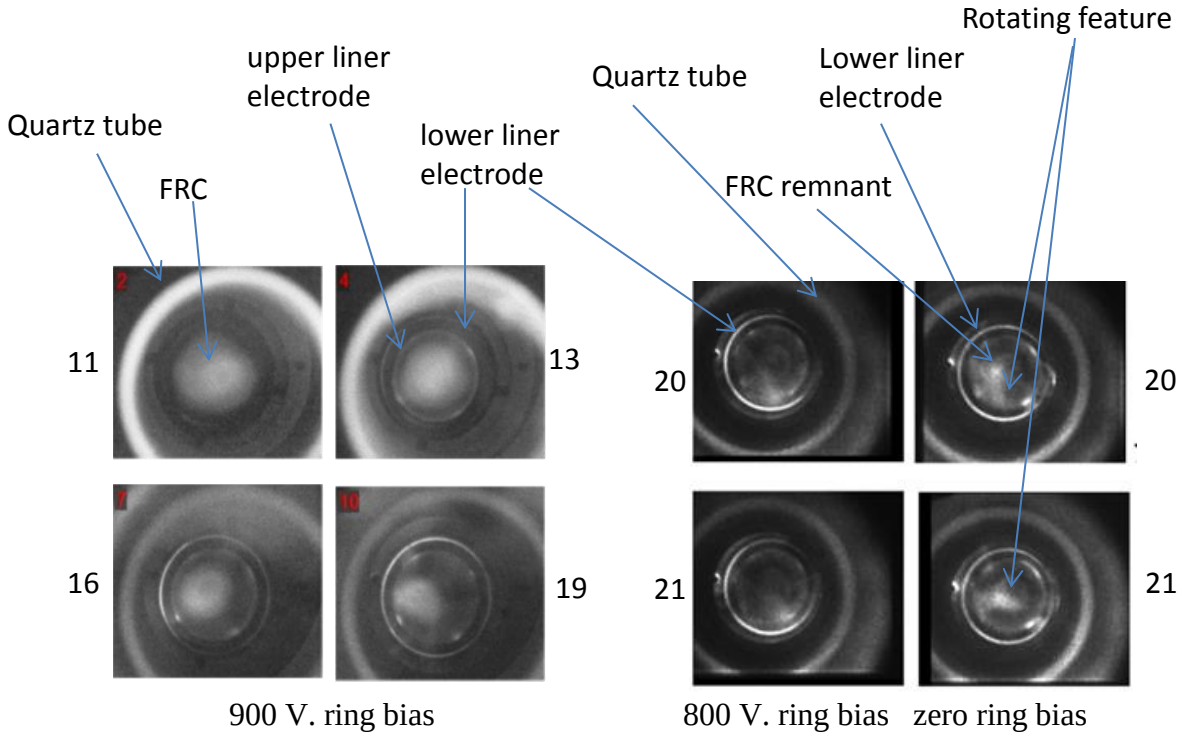


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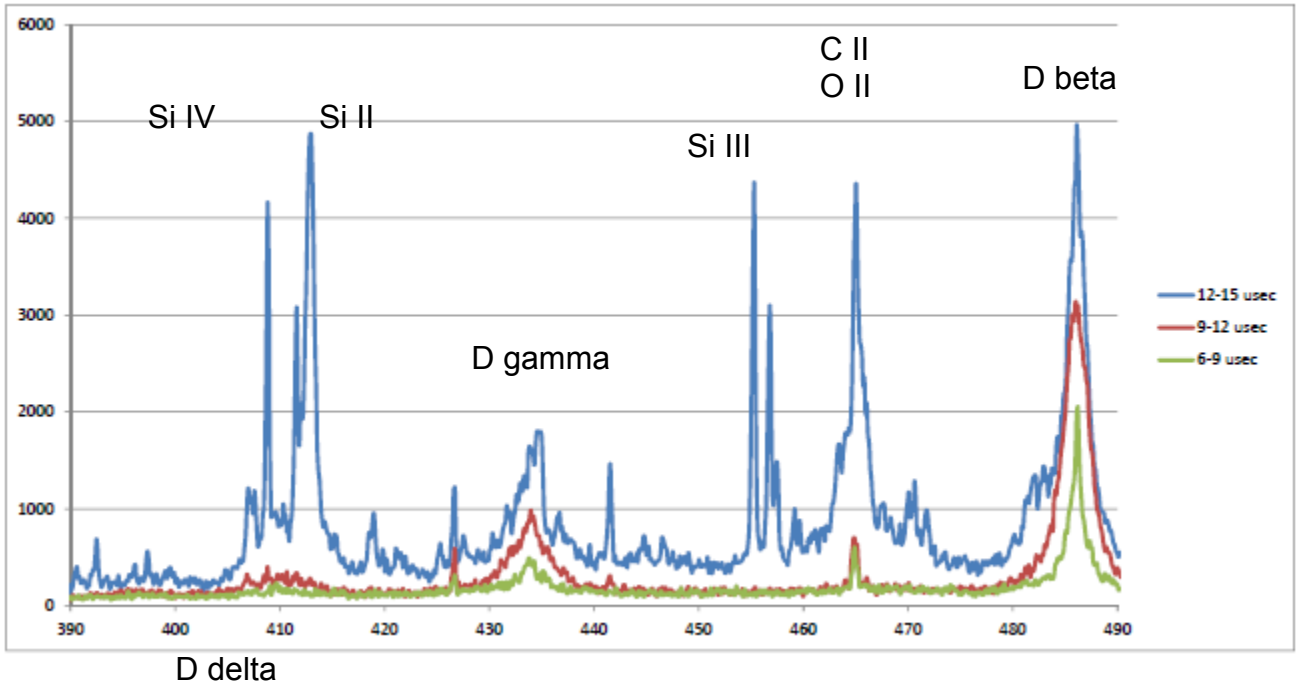


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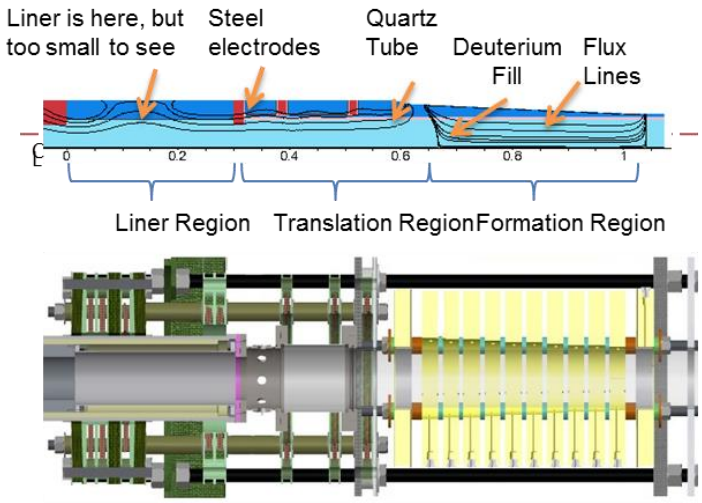


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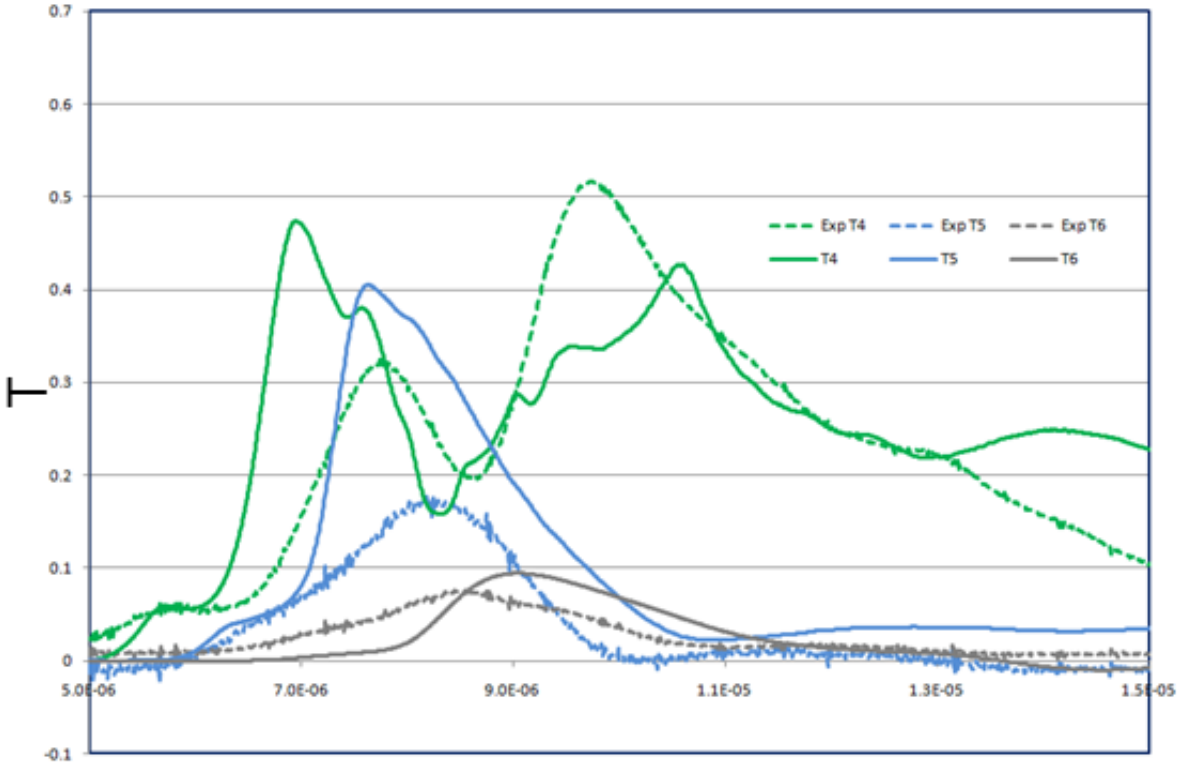


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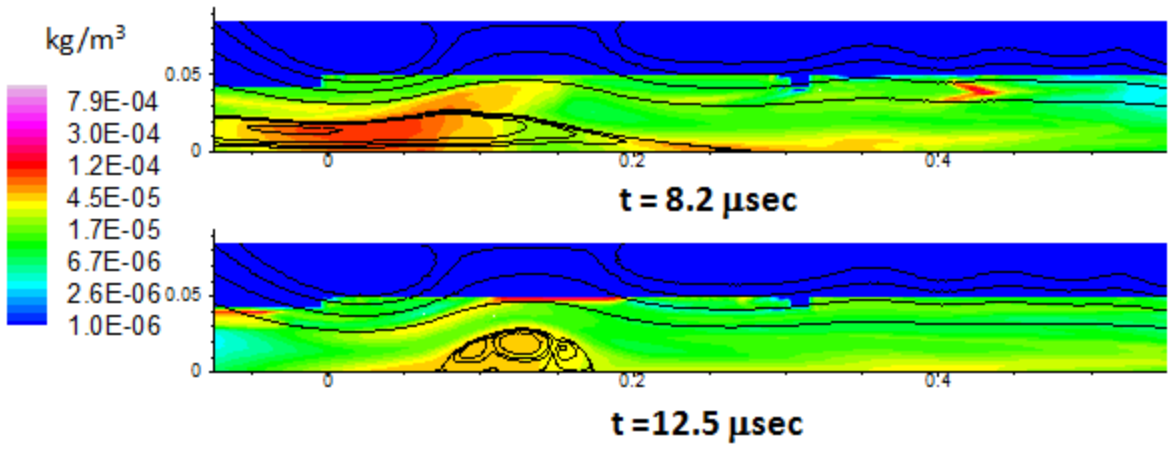


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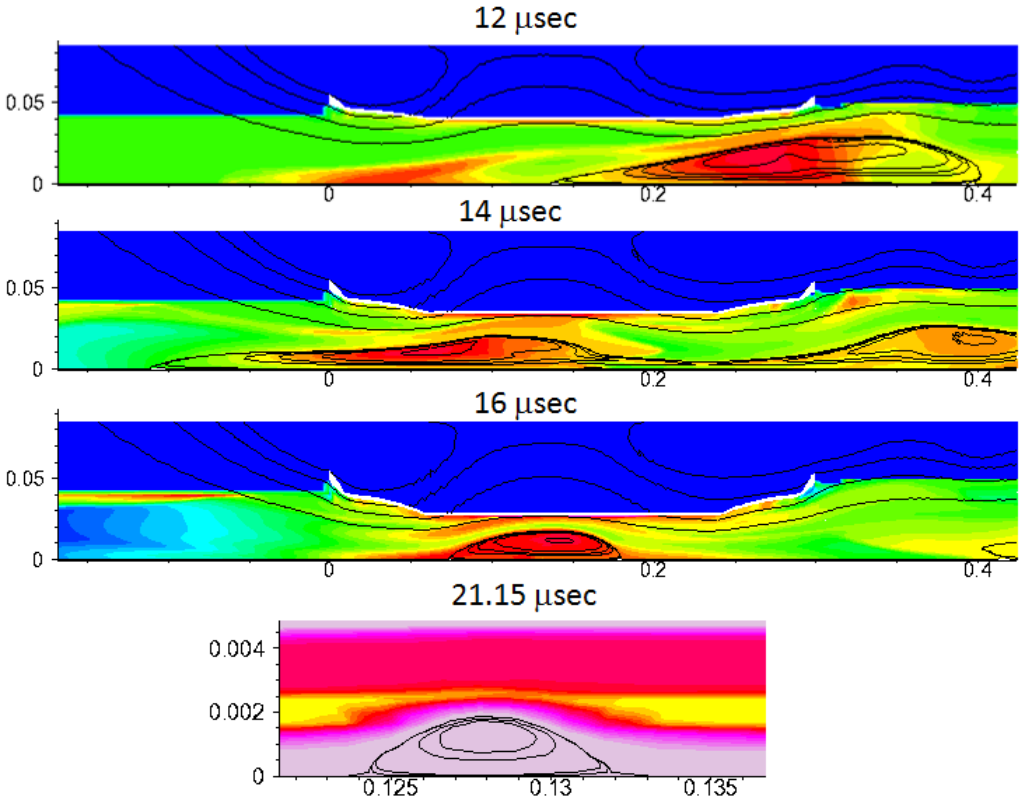


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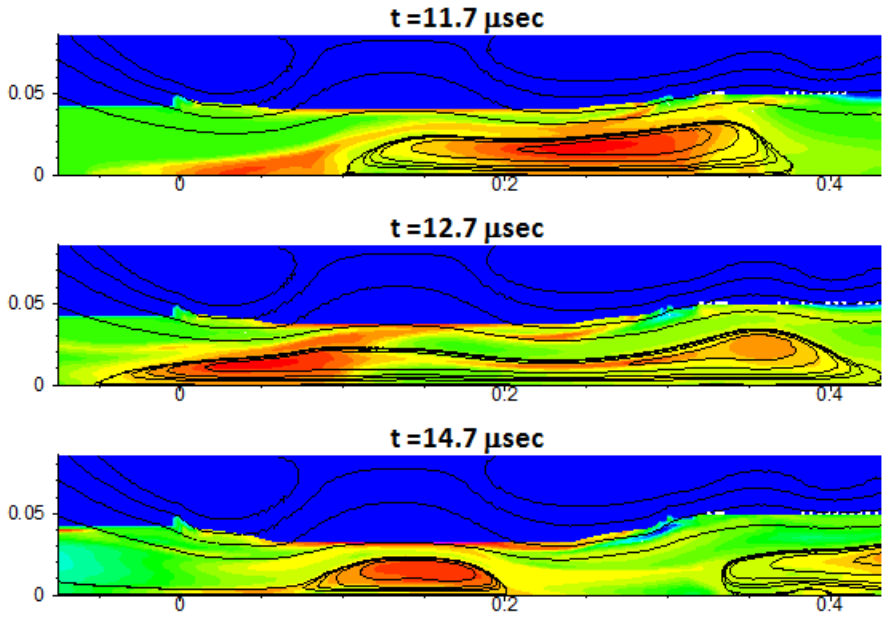


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